AWS ETL Pipeline: Visual Execution Log

Project Architecture Review

This document provides visual validation of the serverless event-driven architecture, documenting the journey of clinical data from raw ingestion to highly available analytical queries.

Phase 1: Secure Ingestion & IAM Provisioning

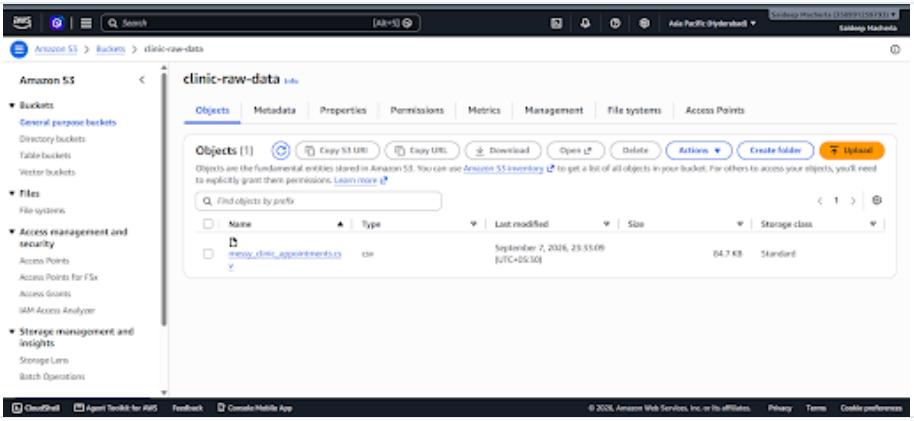


Fig 1.1: S3 Ingestion - `messy_clinic_appointments.csv` uploaded to `clinic-raw-data`

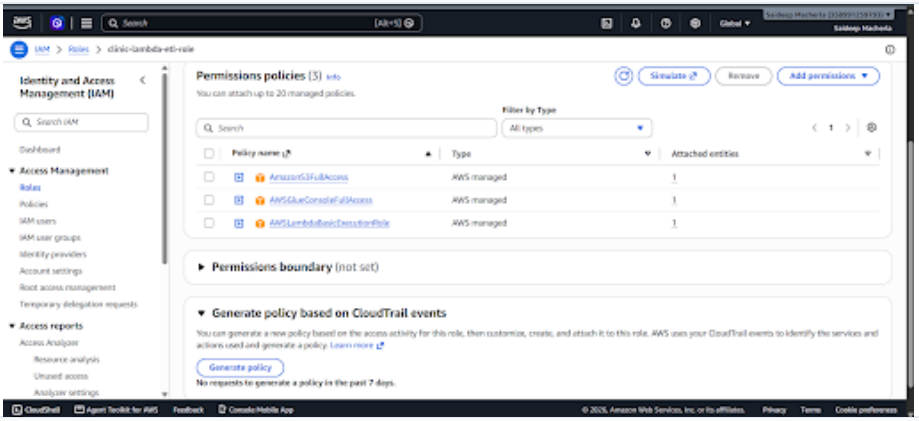


Fig 1.2: IAM Role - `clinic-lambda-etl-role` defining least-privilege policies

> **Execution Nuance:** The pipeline initiates when the raw CSV drops into the S3 bucket. Crucially, the compute engine operates under strict IAM boundaries. The `clinic-lambda-etl-role` ensures the Lambda function possesses `AmazonS3FullAccess` to fetch and overwrite data, alongside `AWSLambdaBasicExecutionRole` to emit diagnostic logs to CloudWatch.

Phase 2: Serverless Compute & Transformation

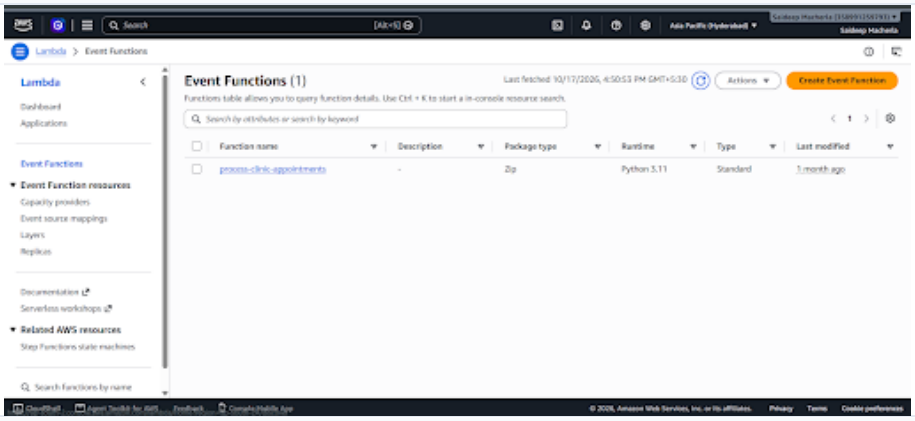


Fig 2.1: Compute Provisioning - `process-clinic-appointments` Lambda (Python 3.11)

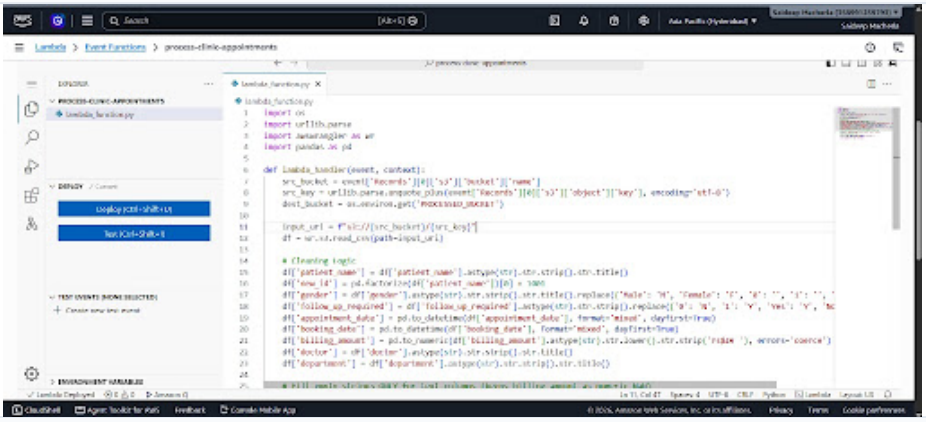


Fig 2.2: ETL Logic - Pandas and awswrangler transformations applied to raw data

Execution Nuance: The `ObjectCreated` event awakens the Lambda function. Inside, the code leverages the AWS Data Wrangler library to pull the S3 URI into memory. The pandas script applies heavy transformations: normalizing text strings to title case, generating deterministic primary keys (`new_id`), parsing mixed date formats, and coercing invalid currency data into strict numeric types.

Phase 3: Idempotent Storage Routing

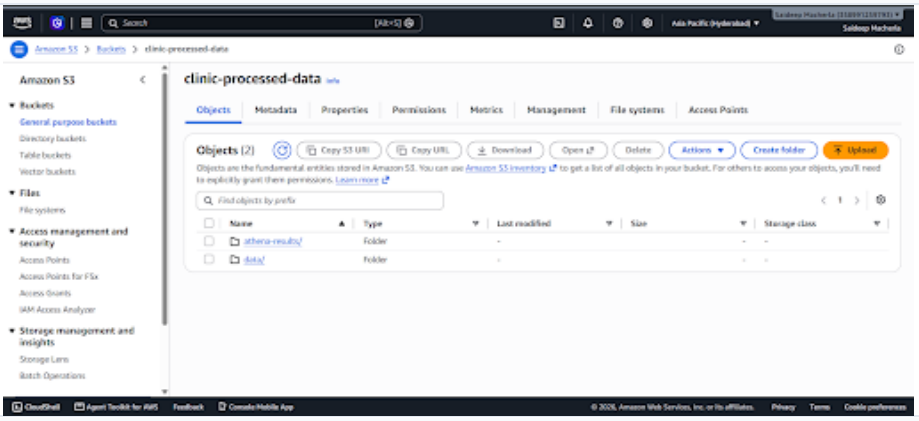


Fig 3.1: Data Governance - Separation of `/data` and `/athena-results` directories

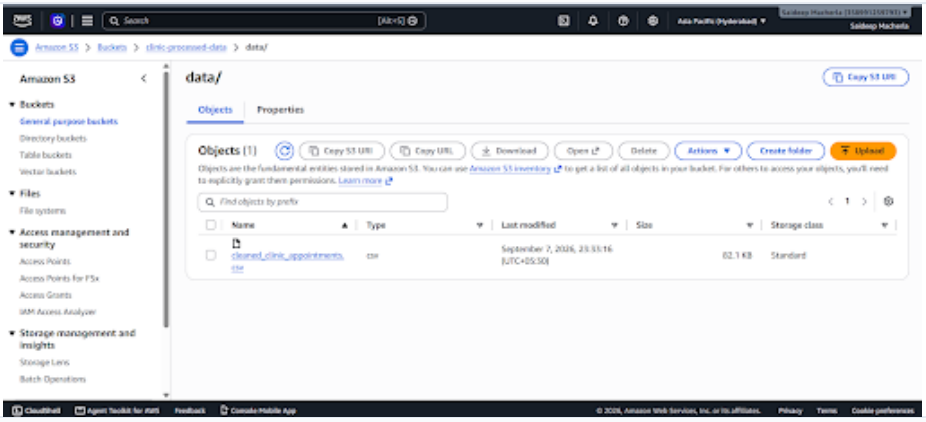


Fig 3.2: Deterministic Load - `cleaned_clinic_appointments.csv` overwrites safely

> **Execution Nuance:** A common failure point in cloud ETL is "data stacking" upon re-execution. This pipeline solves this by defining a fixed output URI. The Lambda function overwrites `cleaned_clinic_appointments.csv` exactly inside the `/data/` folder. This architectural split ensures Athena queries the `/data/` folder exclusively, ignoring its own generated `/athena-results/` cache.

Phase 4: Schema Definition & SQL Analytics

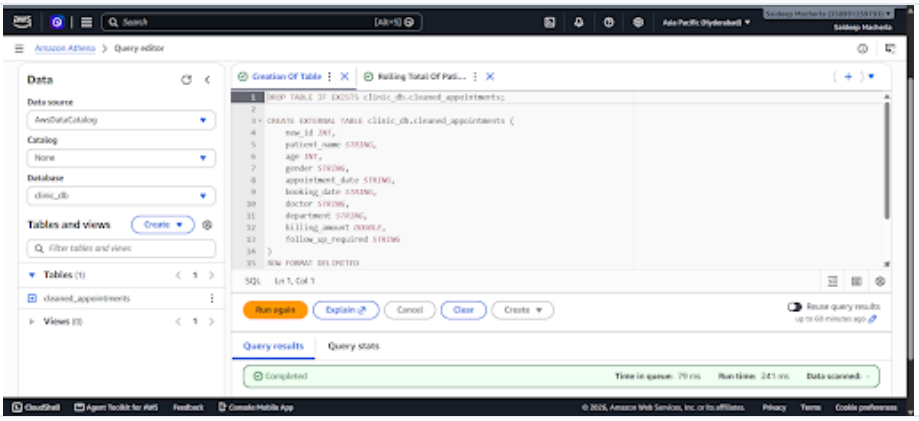


Fig 4.1: Schema Enforcement - DDL execution using external table definitions

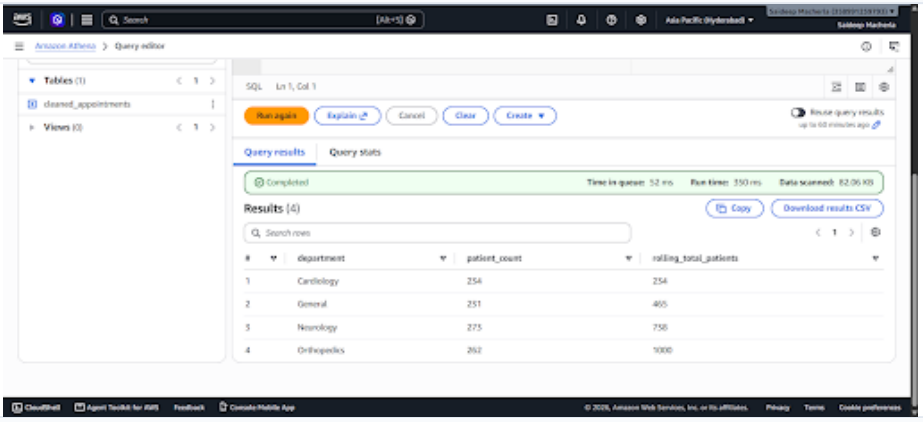


Fig 4.2: Distributed Analytics - Successful execution of SQL window functions

> **Execution Nuance:** To counter `NumberFormatException` crashes caused by empty numeric strings in standard CSVs, the table was created using the `ROW FORMAT DELIMITED` property (LazySimpleSerDe). This allows Athena to gracefully convert missing data to `NULL`. The final query perfectly processes rolling patient totals by department directly over the S3 data lake.